

# MAE 1001

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## Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2025/>

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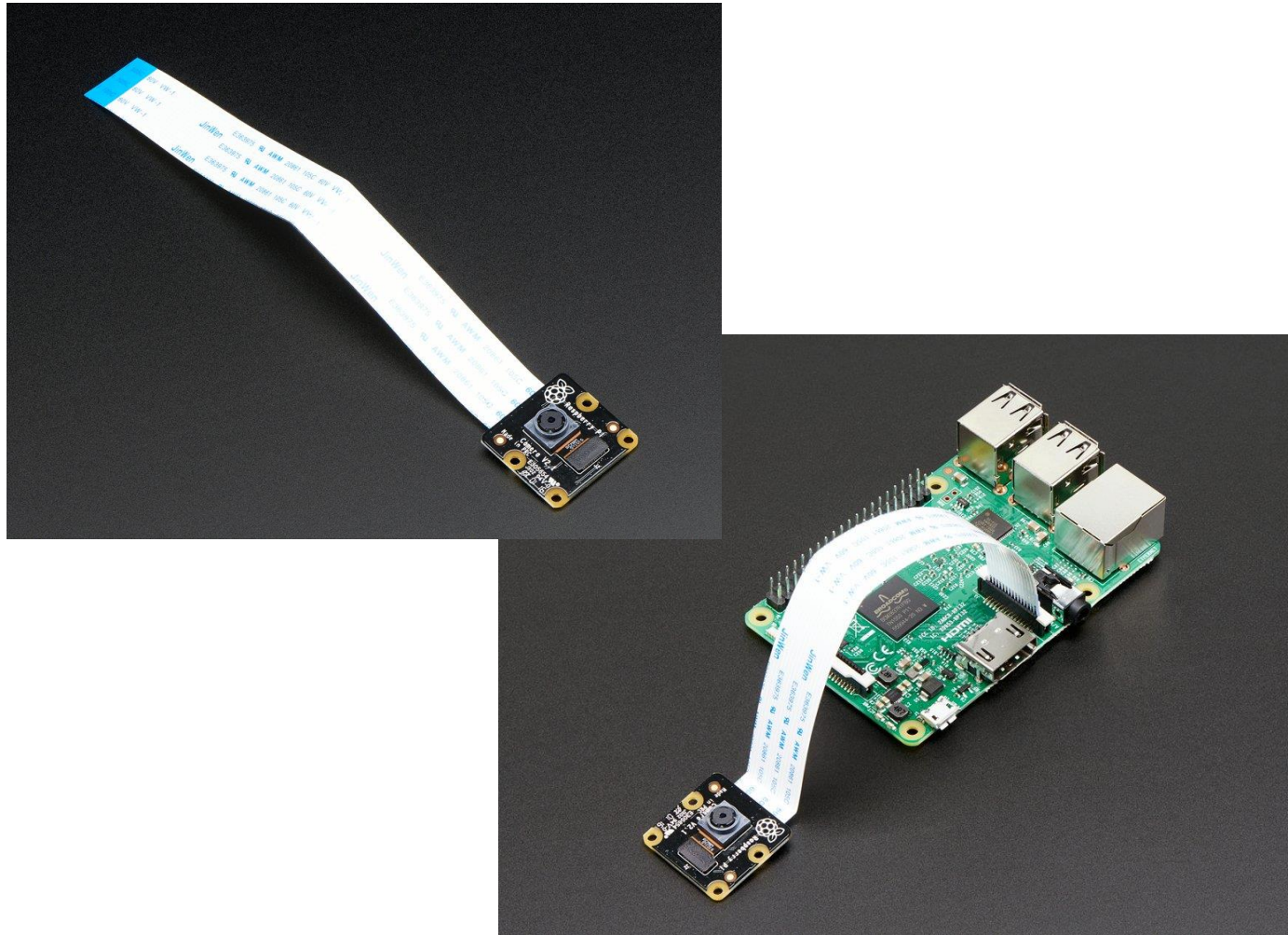
School of Engineering  
& Applied Science

THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu

# Introducing the Pi NoIR Camera

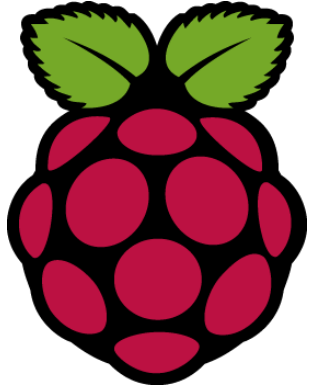


- 8 megapixel native resolution high quality Sony IMX219 image sensor
- 3280 x 2464 pixel static images
- Capture video at
  - 1920 x 1080 p30
  - 1280 x 720 p60
  - 640 x 480 p90 resolutions
- No Infrared (NoIR) filter
  - Infrared photographs or photographing objects in low light (twilight) conditions

Source:

<https://www.adafruit.com/product/3100#description>





### The Picamera2 Library

A libcamera-based Python library  
for Raspberry Pi cameras



Raspberry Pi hardware and Camera hardware interactions  
Low-level image acquisition and processing

## Computer vision on the Raspberry Pi



## The Picamera2 Library



```
# import the necessary packages
```

```
from picamera2 import Picamera2
from picamera2 import Preview
import cv2
import time
import numpy
```

```
# initialize the camera and grab a reference
# to the raw camera capture
# create an instance of the Picamera2
```

```
picam2 = Picamera2()
picam2.start_preview(Preview.QT)
```

```
# set and load the configuration to be used
```

```
capture_config = picam2.create_still_configuration(lores={"size": (320, 240)}, display="lores")
```

```
# start the camera and display a preview
```

```
picam2.start(show_preview=True)
```

```
# buffer time to allow for people to say cheese
time.sleep(0.5)
```

```
# Take the picture and save it as "image.jpg" to the current directory
image = picam2.switch_mode_and_capture_file(capture_config, "image.jpg")
array = picam2.capture_array("main")
```

```
picam2.stop_preview()
```

```
#Turn the camera and preview off
picam2.stop()
```

```
# Create a window to be used (NOT using the WINDOW_AUTOSIZE)
cv2.namedWindow('New Image',cv2.WINDOW_NORMAL)
cv2.resizeWindow('New Image', 600,400) #Resize the window to the desired size
cv2.imshow('New Image', array) #Display the image in the window
cv2.waitKey(0)
```



```
#Exit the program and also close the preview window
```

```
exit()
```



# Goal of the lab segment

## Co-work

- Observe, ask and try in groups

## Make

- Build-a-hack
- Use Pi NoIR Camera to acquire an images
- import OpenCV library

Perform basic image processing functions using OpenCV



## The Picamera2 Library



```
3 import cv2
4 import numpy as np
5
6 #IMAGE COLOR FILTERING
7
8 img = cv2.imread('/home/pi/Desktop/Fall2022/Week4-OpenCV_Camera/savedI
9
10 retval, threshold = cv2.threshold(img, 12, 255, cv2.THRESH_BINARY)
11
12 grayscale = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
13
14 retval2, threshold2 = cv2.threshold(grayscale, 12, 255, cv2.THRESH_BINARY)
15
16 gaus = cv2.adaptiveThreshold(grayscale, 255, cv2.THRESH_BINARY, cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_OTSU)
17
18 retval2, otsu = cv2.threshold(grayscale, 125, 255, cv2.THRESH_BINARY)
19
20 cv2.imshow('original', img)
21 cv2.imshow('threshold', threshold)
22 cv2.imshow('threshold2', threshold2)
23 cv2.imshow('gaus', gaus)
24 cv2.imshow('otsu', otsu)
25 cv2.waitKey(0)
26 cv2.destroyAllWindows()
```

```
Shell x
===== RESTART =====
>>> %Run picamera_opencv_test1.py
===== RESTART =====
>>> %Run picamera_opencv_test7.py
```

